

1st REVIEW

TICKET BOOKING SYSTEM

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Under the Guidance
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GROUP NUMBER:7

Name of Candidate	Role
RAYANKI HIMAVARSHINI	Team Lead & Backend / Database Developer – Java , spring Boot.
GURRAPU NIKSHITHA	Frontend Developer – React , HTML , CSS
VEMUA NAGA VENKATA SAI CHANDU	Concurrency & Integration developer.

ABSTRACT

- Web application that simplifies and automates ticket booking for movies, events and entertainment services.
- Users register, log in, browse shows, select date, time and seats, pay online, and receive a unique booking ID and digital ticket.
- Users can view booking history and cancel tickets; administrators manage shows, seats, users, bookings and reports.
- Built with React (frontend), Java and Spring Boot (backend) and MySQL (users, shows, seats, bookings and payments).
- Redis distributed locking temporarily locks selected seats to prevent double booking; RabbitMQ queues booking and payment requests for better performance.
- Supports cancellation and refund processing for a secure, reliable, scalable and concurrency-safe experience.

INTRODUCTION

- A web-based application that makes ticket booking simple, fast and convenient.
- Users register, log in, browse movies and events, choose dates, times and seats, and complete the booking online.
- After successful payment, users receive a unique booking ID and a digital ticket.
- React is used for the frontend, Java and Spring Boot for backend services, and MySQL for storing application data.
- Redis locks selected seats during booking, and RabbitMQ handles requests during high traffic.
- Includes cancellation, refunds, booking history and administrative features for managing shows, seats, users and bookings.

EXISTING SYSTEM

- Manual or basic online ticket booking
- Limited real-time seat availability
- Risk of double booking and seat overselling
- Slow processing during high traffic
- Poor concurrency handling
- Difficult cancellation and refund process
- Limited scalability
- Difficult management of users, shows, seats, and bookings
- Limited booking and reporting features

PROBLEM

- Traditional systems struggle when many users try to book at the same time.
- Two users may try the same seat at the same moment, causing double booking or overselling.
- Manual processes are slow and give limited information about shows, seats and booking status.
- Cancellations, refunds, payments and booking records create extra administrative work.
- A system is needed with real-time seat availability, secure booking, efficient payment processing and proper cancellation management.
- Each seat must be booked by only one user, even under high concurrency; this project solves it with Redis seat locking and RabbitMQ.

SOLUTION(Contribution)

- An online platform with real-time seat availability, secure booking, payment processing and digital tickets.
- Redis distributed locking temporarily locks a selected seat so no other user can book it.
- RabbitMQ queues booking and payment requests so the system handles high traffic efficiently.
- Supports booking history, cancellation and refund processing.
- Administrators manage shows, seats, users and bookings, and generate reports.
- Reduces manual work, prevents double booking and overselling, and improves the user experience.

Objectives

- To develop a simple and user-friendly online ticket booking system.
- To allow users to view shows, check seat availability, select seats, and book tickets online.
- To prevent double booking and seat overselling using Redis distributed locking.
- To handle multiple booking requests efficiently using RabbitMQ.
- To provide secure payment, cancellation, refund, booking history, and digital ticket features.

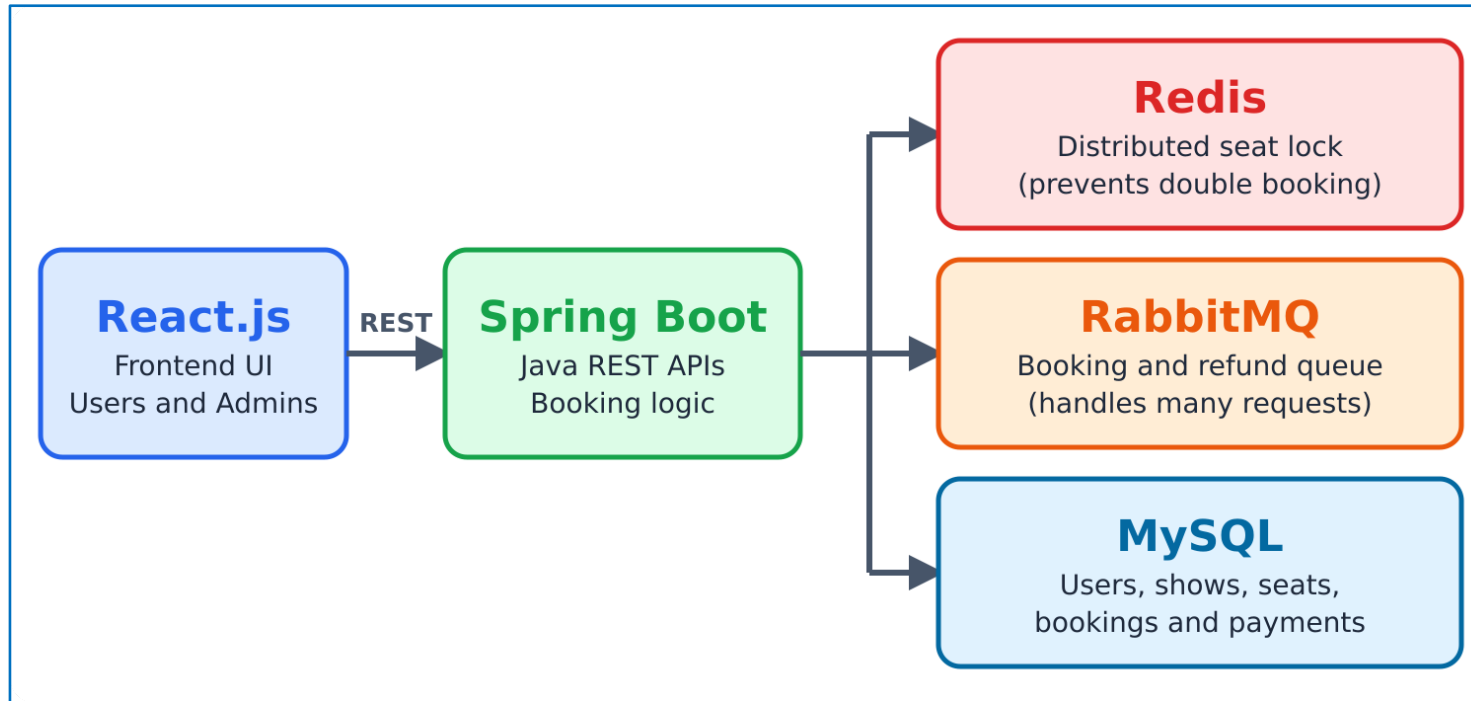
Modules

- **User Module** – Registration, login, profile management, and booking history.
- **Show & Seat Module** – Browse movies/events, view show timings and ticket prices, and check real-time seat availability.
- **Booking & Payment Module** – Select seats, book tickets, process secure payments, and generate digital tickets.
- **Concurrency & Queue Module** – Use Redis for distributed seat locking to prevent double booking and RabbitMQ to efficiently handle multiple booking requests during high traffic.
- **Admin Module** – Manage shows, seats, users, bookings, cancellations, refunds, reports, and analytics.

SOFTWARE REQUIREMENTS

- **Frontend:** React.js
- **Backend:** Java & Spring Boot
- **Database:** MySQL
- **Seat Locking:** Redis
- **Message Queue:** RabbitMQ
- **Operating System:** Windows / Linux
- **Development Environment:** VS Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Microsoft Edge

SYSTEM ARCHITECTURE

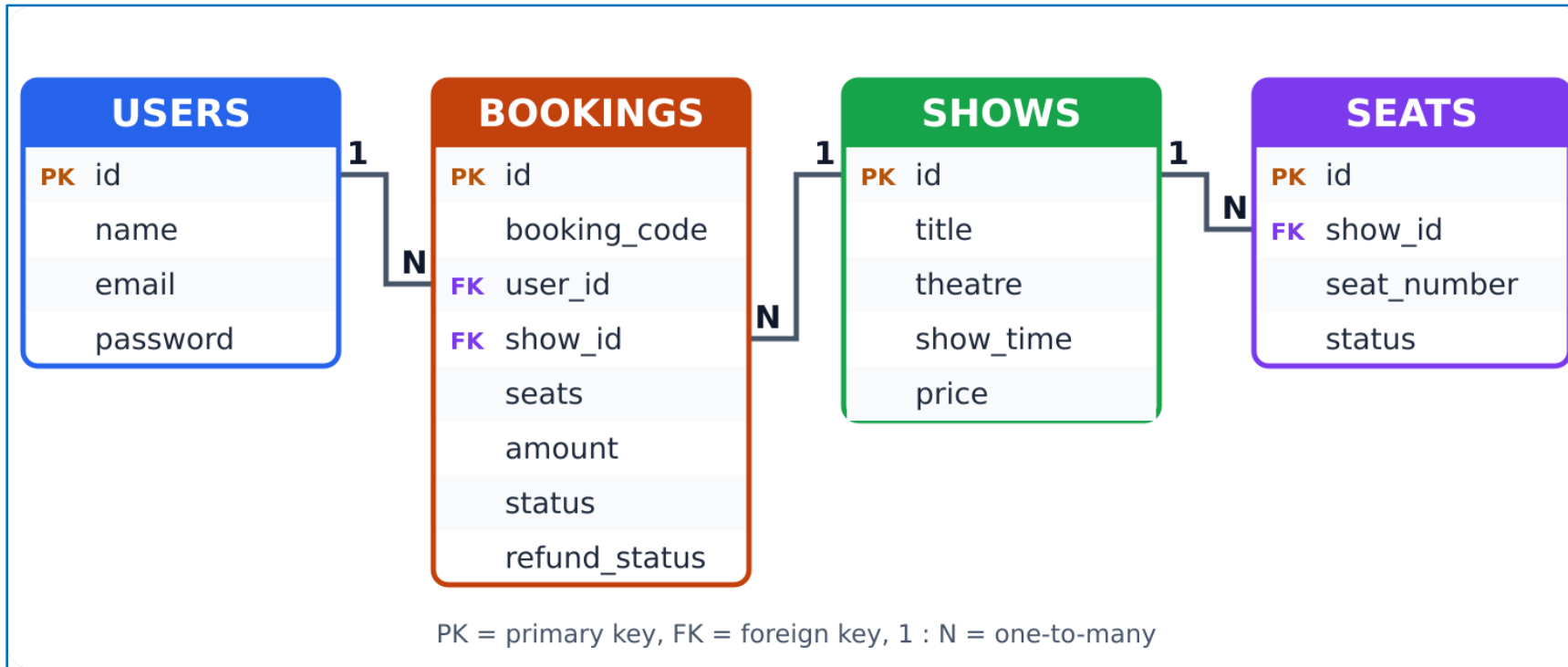


- React sends REST requests
- Spring Boot handles the booking logic
- Redis locks a seat while it is being booked
- RabbitMQ queues booking and refund requests
- MySQL stores users, shows, seats and bookings

BOOKING FLOW

1. User selects a show, date, time and seats.
2. Redis locks the selected seats; only one user can hold a seat.
3. The booking is saved as PENDING and sent to the RabbitMQ queue.
4. Payment is processed, the seats are booked in MySQL and the lock is released.
5. A unique booking ID and digital ticket are generated.
6. Cancelling frees the seats and sends the refund through a queue.

DATABASE DESIGN



One user has many bookings; one show has many bookings and many seats.

APPLICATION SCREENS

localhost:5173 - Select Seats

Select Seats

Avengers: Endgame | PVR Cinemas

A1	A2	A3	A4	A5	A6
B1	B2	B3	B4	B5	B6
C1	C2	C3	C4	C5	C6
D1	D2	D3	D4	D5	D6

Available Selected Locked Booked

Pay & Book - Rs. 400

localhost:5173 - My Bookings

My Bookings

Avengers: Endgame (PVR) **CONFIRMED**

Seats: B3, B4 | Amount: Rs. 400

Booking ID: BK3F9A21C

Cancel

Inception (INOX) **CANCELLED**

Seats: C1 | Amount: Rs. 180

Booking ID: BK7D20E4A

Refund: REFUNDED

Seat selection with live status (left) and My Bookings with refund status (right)

CONCLUSION

- One online platform to browse shows, select seats, pay and receive digital tickets.
- Redis seat locking prevents two users from booking the same seat.
- RabbitMQ queues keep booking and payment requests manageable during high traffic.
- Cancellation, refund and administrative features reduce manual work.
- The result is a secure, reliable, scalable and concurrency-safe ticket booking system.

FUTURE SCOPE

- Mobile application for Android and iOS.
- Cloud deployment for better scalability and availability.
- Advanced booking, revenue and user analytics reports for administrators.
- Email and SMS notifications for bookings and cancellations.
- Secure online payment gateway integration.
- AI-based recommendations for movies and events.
- Support for multiple theatres and locations.
- Stronger authentication and authorization.

THANK YOU